

§1 数学准备

§1.1 矢量的代数运算.

点积: $\vec{A} \cdot \vec{B} = A_i B_j \delta_{ij}$

叉积: $\vec{A} \times \vec{B} = \epsilon_{ijk} e_i A_j B_k$

* 证明: $\epsilon_{ijk} \epsilon_{lmn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}$

证: $LHS = \epsilon_{ijk} \epsilon_{lmn} \delta_{il}$
 $= e_i (e_j \times e_k) \cdot e_i (e_m \times e_n)$
 $= (e_j \times e_k) \cdot (e_m \times e_n)$

① $(j, k) = (m, n) / (n, m)$ 则 $LHS = 1$

② $(j, k) = (m, n)$

$LHS = 1 = RHS$

③ $(j, k) = (n, m)$

$LHS = -1 = RHS$

(1) $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B})$

证: $\vec{A} \times (\vec{B} \times \vec{C}) = \epsilon_{ijk} e_i A_j (\vec{B} \times \vec{C})_k$
 $= \epsilon_{ijk} e_i A_j \epsilon_{kmn} B_m C_n$
 $= \epsilon_{kij} \epsilon_{kmn} e_i A_j B_m C_n$
 $= (\delta_{im} \delta_{jn} - \delta_{jm} \delta_{in}) e_i A_j B_m C_n$
 $= \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B}) \quad \square$

(2) $\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$

证: $\vec{A} \cdot (\vec{B} \times \vec{C}) = \epsilon_{ijk} A_i B_j C_k = \epsilon_{ikl} A_i B_k C_l \sim \epsilon_{ikl}$

ϵ_{ikl} 具有“循环”性质 $\square$

§ 1.2 矢量的导数运算

1. 梯度. $\nabla\varphi = \frac{\partial\varphi}{\partial x_i} \mathbf{e}_i = \nabla\varphi \cdot \mathbf{e}_i$, $\nabla\varphi := \frac{\partial\varphi}{\partial x_i}$, $\nabla = \mathbf{e}_i \partial_i$

2. 散度.

$$\nabla \cdot \vec{A} = (\mathbf{e}_i \partial_i) \cdot (A_j \mathbf{e}_j) = \delta_{ij} \partial_i A_j = \partial_i A_i$$

$$\nabla \cdot \vec{A} = 0 \rightarrow \vec{A} \text{ 是无源场}$$

$$\text{点电荷 } \vec{E}(\vec{r}) = \frac{\vec{r}}{r^3}$$

$$\nabla \cdot \vec{E} = \mathbf{e}_i \partial_i \delta_{ij} E_j = \partial_i E_i = \partial_i \left(\frac{r_i}{r^3} \right)$$

$$= \frac{\partial r_i}{\partial r^3} = 3r_i \frac{\partial r}{\partial r^4} = \frac{3}{r^3} - 3r_i \cdot \frac{r_i}{r^5} = 0 \quad (r \neq 0) \quad \partial_i r = \partial_i r \mathbf{e}_i = \mathbf{e}_i = \frac{r_i}{r}$$

3. 旋度

$$\nabla \times \vec{A} = \epsilon_{ijk} \mathbf{e}_i \partial_j A_k$$

$$\nabla \times \nabla\varphi = \epsilon_{ijk} \mathbf{e}_i \partial_j \partial_k \varphi = 0 \quad j, k \text{ 互换后 } (i, j, k), (i, k, j) \text{ 抵消.}$$

$$\nabla \cdot (\nabla \times \vec{A}) = \mathbf{e}_i \partial_i \delta_{ij} \epsilon_{jkl} \mathbf{e}_j \partial_k A_l = \epsilon_{ikl} \partial_i \partial_k A_l =$$

$$= -\epsilon_{kil} \partial_k \partial_i A_l \rightarrow \nabla \cdot (\nabla \times \vec{A}) = 0$$

4. 复合运算法则.

$$\nabla(a\varphi + b\psi) = a\nabla\varphi + b\nabla\psi$$

$$\nabla \cdot (a\vec{A} + b\vec{B}) = a\nabla \cdot \vec{A} + b\nabla \cdot \vec{B}$$

$$\nabla(\varphi\psi) = \varphi\nabla\psi + \psi\nabla\varphi$$

$$\nabla \cdot (\varphi\vec{A}) = \varphi\nabla \cdot \vec{A} + \varphi_{,i} A_i$$

$$\nabla \times (\varphi\vec{A}) = \varphi\nabla \times \vec{A} + \varphi(\nabla \times \vec{A})$$

$$\nabla \cdot (\vec{A} \times \vec{B}) = \nabla_A \cdot (\vec{B} \times \vec{C}) + \nabla_B \cdot (\vec{A} \times \vec{C})$$

$$= (\nabla_A \times \vec{A}) \cdot \vec{B} - (\nabla_A \times \vec{B}) \cdot \vec{A}$$

$$= (\nabla \times \vec{A}) \cdot \vec{B} - (\nabla \times \vec{B}) \cdot \vec{A}$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$$

$$\nabla \times (\vec{A} \times \vec{B}) = \nabla_A \times (\vec{B} \times \vec{C}) + \nabla_B \times (\vec{A} \times \vec{C})$$

$$= \vec{A}(\nabla_B \cdot \vec{C}) - \vec{C}(\nabla_B \cdot \vec{A}) + \vec{B}(\nabla_A \cdot \vec{C}) - \vec{C}(\nabla_A \cdot \vec{B})$$

$$= (\vec{B} \cdot \nabla) \vec{A} - (\nabla \cdot \vec{A}) \vec{B} + (\nabla \cdot \vec{B}) \vec{A} - (\vec{A} \cdot \nabla) \vec{B}$$

思考: 1. $\vec{A} \times \vec{B} = \epsilon_{ijk} \epsilon_i A_j B_k$ $\epsilon_{ijk} \epsilon_{lmn} = \delta_{jl} \delta_{kn} - \delta_{jn} \delta_{kl}$

证明 $\nabla \times (\vec{A} \times \vec{B}) = \dots$

2. $\vec{E} = \vec{0}$ 为常 分析 $\nabla \times (\vec{A} \times \vec{B})$

$$\begin{aligned} \nabla \times (\vec{A} \times \vec{B}) &= \vec{e}_i \partial_i \times \epsilon_{jmn} \vec{e}_j A_m B_n \\ &= \epsilon_{kij} \epsilon_{jmn} \vec{e}_k \partial_i A_m B_n = \epsilon_{jki} \epsilon_{jmn} \vec{e}_k \partial_i (A_m B_n) \\ &= (\delta_{km} \delta_{in} - \delta_{kn} \delta_{im}) \vec{e}_k [\partial_i A_m B_n + A_m \partial_i B_n] \\ &= (\vec{A} \cdot \nabla) \vec{B} - (\nabla \cdot \vec{A}) \vec{B} + (\nabla \cdot \vec{B}) \vec{A} - (\vec{B} \cdot \nabla) \vec{A} \end{aligned}$$

$$\begin{aligned} \nabla \times (\vec{A} \times \vec{B}) &= (\vec{A} \cdot \nabla) \vec{B} - (\nabla \cdot \vec{A}) \vec{B} + (\nabla \cdot \vec{B}) \vec{A} - (\vec{B} \cdot \nabla) \vec{A} \\ &= -(\nabla \cdot \vec{A}) \vec{B} - (\vec{B} \cdot \nabla) \vec{A} \end{aligned}$$

$$\begin{aligned} \vec{A} \times (\nabla \times \vec{B}) &= \epsilon_{ijk} \epsilon_i A_j \epsilon_{kmn} \partial_m B_n = \epsilon_{kij} \epsilon_{kmn} \epsilon_i A_j \partial_m B_n \\ &= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \epsilon_i A_j \partial_m B_n = A_j \epsilon_i \partial_i B_j - A_j \partial_j B_i \\ &= \underline{\nabla_n (\vec{A} \cdot \vec{B})} - (\vec{A} \cdot \nabla) \vec{B} \quad * \quad \partial_m B_n, \text{ 为标量 } B \rightarrow \partial_n \end{aligned}$$

$$\hookrightarrow \nabla_n (\vec{A} \cdot \vec{B}) = (\vec{B} \cdot \nabla) \vec{A} + \vec{B} \times (\nabla \times \vec{A})$$

$$\nabla_n (\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \nabla) \vec{B} + \vec{A} \times (\nabla \times \vec{B})$$

$$\begin{aligned} \nabla (\vec{A} \cdot \vec{B}) &= \nabla_n (\vec{A} \cdot \vec{B}) + \nabla_B (\vec{A} \cdot \vec{B}) \\ &= (\vec{B} \cdot \nabla) \vec{A} + \vec{B} \times (\nabla \times \vec{A}) + (\vec{A} \cdot \nabla) \vec{B} + \vec{A} \times (\nabla \times \vec{B}) \end{aligned}$$

$$\hookrightarrow \nabla (\vec{A} \cdot \vec{A}) = \nabla \vec{A}^2 = 2 [(\vec{A} \cdot \nabla) \vec{A} + \vec{A} \times (\nabla \times \vec{A})]$$

例: $\nabla \times \vec{E}_0 \sin(\vec{K} \cdot \vec{r})$

$$\begin{aligned} &= [\nabla (\vec{K} \cdot \vec{r})] \times \vec{E}_0 \cos(\vec{K} \cdot \vec{r}) \\ &= [(\vec{K} \cdot \nabla) \vec{r} + \vec{K} \times (\nabla \times \vec{r}) + (\vec{r} \cdot \nabla) \vec{K} + \vec{r} \times (\nabla \times \vec{K})] \times \vec{E}_0 \cos(\vec{K} \cdot \vec{r}) \\ &= \vec{K} \times \vec{E}_0 \cos(\vec{K} \cdot \vec{r}) \end{aligned}$$

$$\nabla \times \vec{r} = \epsilon_{ijk} \epsilon_i \partial_j r_k = \epsilon_{ijk} \epsilon_i \delta_{jk} = -\epsilon_{ikj} \epsilon_i \delta_{kj} \rightarrow \nabla \times \vec{r} = \vec{0}$$

$$(\vec{K} \cdot \nabla) \vec{r} = k_i \partial_i \epsilon_j r_j = k_i \epsilon_j \delta_{ij} = k_i \epsilon_i = \vec{K}$$

$$\begin{aligned}\nabla \times (\vec{E}_0 e^{i(\vec{k} \cdot \vec{r})}) &= \nabla e^{i(\vec{k} \cdot \vec{r})} \times \vec{E}_0 = i \cdot \nabla(\vec{k} \cdot \vec{r}) e^{i(\vec{k} \cdot \vec{r})} \times \vec{E}_0 \\ &= i\vec{k} \times \vec{E}_0 e^{i(\vec{k} \cdot \vec{r})}\end{aligned}$$

$$\begin{aligned}\nabla \cdot (\vec{E}_0 e^{i(\vec{k} \cdot \vec{r})}) &= i \nabla(\vec{k} \cdot \vec{r}) \cdot \vec{E}_0 e^{i(\vec{k} \cdot \vec{r})} \\ &= i\vec{k} \cdot \vec{E}_0 e^{i(\vec{k} \cdot \vec{r})}\end{aligned}$$

思考: $\vec{A} < (\nabla \times \vec{B})$, $\vec{B} < \nabla \times \vec{A}$ 有何差异.

6. 二阶导数:

$$\nabla \cdot (\nabla \varphi) = (\partial_i \partial_i) (\partial_j \partial_j) \varphi = \nabla^2 \varphi$$

$$\begin{aligned}\nabla^2 (\varphi \psi) &= \nabla \cdot (\varphi \nabla \psi + \psi \nabla \varphi) \\ &= \varphi \nabla^2 \psi + \psi \nabla^2 \varphi + 2 \nabla \varphi \cdot \nabla \psi\end{aligned}$$

例: 证明 $\nabla \cdot (\nabla \times \vec{A}) = 0$

$$\begin{aligned}\nabla \cdot (\nabla \times \vec{A}) &= \partial_i \epsilon_{ijk} \partial_j A_k = \epsilon_{ijk} \partial_i \partial_j A_k \\ &= -\epsilon_{jki} \partial_j \partial_i A_k \rightarrow \nabla \cdot (\nabla \times \vec{A}) = 0\end{aligned}$$

证明 $\nabla \times (\nabla \varphi) = 0$

$$\nabla \times (\nabla \varphi) = \epsilon_{ijk} \partial_i \partial_j \partial_k \varphi = -\epsilon_{ikj} \partial_i \partial_k \partial_j \varphi = 0$$

例: 证明 $\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$

$$\begin{aligned}\nabla \times (\nabla \times \vec{E}) &= \epsilon_{ijk} \partial_i \partial_j \epsilon_{kmn} \partial_m E_n = \epsilon_{kij} \epsilon_{kmn} \partial_i \partial_j \partial_m E_n \\ &= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \partial_i \partial_j \partial_m E_n \\ &= \partial_i \partial_k \partial_j E_j - \partial_j \partial_j (\partial_i E_i) \\ &= \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}\end{aligned}$$

6. 柱坐标下的导数运算.

$$d\vec{r} = dr \vec{e}_r + r d\phi \vec{e}_\phi + dz \vec{e}_z$$

$$dV = r^2 d\phi dz$$

$$\nabla = \vec{e}_x \partial_x + \vec{e}_y \partial_y + \vec{e}_z \partial_z$$

$$= \vec{e}_r \partial_r + \vec{e}_\phi \frac{1}{r} \partial_\phi + \vec{e}_z \partial_z$$

柱坐标参数: $h_1=1$, $h_2=r$, $h_3=1$

$$\nabla \varphi = \vec{e}_r \partial_r \varphi + \vec{e}_\phi \frac{1}{r} \partial_\phi \varphi + \vec{e}_z \partial_z \varphi$$

$$\frac{\partial \vec{e}_r}{\partial r} = 0, \quad \frac{\partial \vec{e}_\phi}{\partial r} = 0, \quad \frac{\partial \vec{e}_z}{\partial r} = 0$$

$$\frac{\partial \vec{e}_r}{\partial \phi} = \vec{e}_\phi, \quad \frac{\partial \vec{e}_\phi}{\partial \phi} = -\vec{e}_r, \quad \frac{\partial \vec{e}_z}{\partial \phi} = 0$$

$$\frac{\partial \vec{e}_r}{\partial z} = 0, \quad \frac{\partial \vec{e}_\phi}{\partial z} = 0, \quad \frac{\partial \vec{e}_z}{\partial z} = 0$$

$$\begin{aligned} \nabla \cdot \vec{A} &= (\vec{e}_r \partial_r + \vec{e}_\phi \frac{1}{r} \partial_\phi + \vec{e}_z \partial_z) \cdot (A_r \vec{e}_r + A_\phi \vec{e}_\phi + A_z \vec{e}_z) \\ &= \vec{e}_r \cdot [\partial_r (A_r \vec{e}_r) + \partial_r (A_\phi \vec{e}_\phi) + \partial_r (A_z \vec{e}_z)] + \vec{e}_\phi \frac{1}{r} [\partial_\phi (A_r \vec{e}_r) + \partial_\phi (A_\phi \vec{e}_\phi) + \partial_\phi (A_z \vec{e}_z)] \\ &\quad + \vec{e}_z [\partial_z (A_r \vec{e}_r) + \partial_z (A_\phi \vec{e}_\phi) + \partial_z (A_z \vec{e}_z)] \\ &= \partial_r A_r + \frac{A_r}{r} + \frac{1}{r} \partial_\phi A_\phi + \partial_z A_z \\ &= \frac{1}{r} \partial_r (r A_r) + \frac{1}{r} \partial_\phi A_\phi + \partial_z A_z \end{aligned}$$

$$\begin{aligned} \nabla \times \vec{A} &= (\vec{e}_r \partial_r + \vec{e}_\phi \frac{1}{r} \partial_\phi + \vec{e}_z \partial_z) \times (A_r \vec{e}_r + A_\phi \vec{e}_\phi + A_z \vec{e}_z) \\ &= \vec{e}_r \times \vec{e}_\phi (\partial_r A_\phi) + \vec{e}_r \times \vec{e}_z (\partial_r A_z) + \frac{1}{r} \vec{e}_\phi \times \vec{e}_r (\partial_\phi A_r) + \frac{A_\phi}{r} \vec{e}_\phi \times (-\vec{e}_r) \\ &\quad \vec{e}_\phi \times \vec{e}_z \frac{1}{r} \partial_\phi A_z + \vec{e}_z \times \vec{e}_r (\partial_z A_r) + \vec{e}_z \times \vec{e}_\phi (\partial_z A_\phi) \\ &= \partial_r A_\phi \vec{e}_z - \partial_r A_z \vec{e}_\phi - \frac{1}{r} \partial_\phi A_r \vec{e}_z + \frac{A_\phi}{r} \vec{e}_z + \frac{1}{r} \partial_\phi A_z \vec{e}_r + \partial_z A_r \vec{e}_\phi - \partial_z A_\phi \vec{e}_r \\ &= \left(\frac{1}{r} \partial_\phi A_z - \partial_z A_\phi \right) \vec{e}_r + \left(\partial_z A_r - \partial_r A_z \right) \vec{e}_\phi + \left(\partial_r A_\phi - \frac{1}{r} \partial_\phi A_r + \frac{A_\phi}{r} \right) \vec{e}_z \\ &= \left(\frac{1}{r} \partial_\phi A_z - \partial_z A_\phi \right) \vec{e}_r + \left(\partial_z A_r - \partial_r A_z \right) \vec{e}_\phi + \left(\frac{1}{r} \partial_r (r A_\phi) - \frac{1}{r} \partial_\phi A_r \right) \vec{e}_z \end{aligned}$$

$$\begin{aligned} \nabla^2 \varphi &= (\nabla \cdot \nabla) \varphi = \nabla \cdot \nabla \varphi \\ &= (\vec{e}_r \partial_r + \vec{e}_\phi \frac{1}{r} \partial_\phi + \vec{e}_z \partial_z) \cdot (\vec{e}_r \partial_r \varphi + \vec{e}_\phi \frac{1}{r} \partial_\phi \varphi + \vec{e}_z \partial_z \varphi) \\ &= \vec{e}_r \cdot [\partial_r (\vec{e}_r \partial_r \varphi) + \partial_r (\vec{e}_\phi \frac{1}{r} \partial_\phi \varphi) + \partial_r (\vec{e}_z \partial_z \varphi)] + \vec{e}_\phi \cdot \frac{1}{r} [\partial_\phi (\vec{e}_r \partial_r \varphi) + \partial_\phi (\vec{e}_\phi \frac{1}{r} \partial_\phi \varphi) + \\ &\quad + \vec{e}_z \cdot [\partial_z (\vec{e}_r \partial_r \varphi) + \partial_z (\vec{e}_\phi \frac{1}{r} \partial_\phi \varphi) + \partial_z (\vec{e}_z \partial_z \varphi)] \quad \partial_\phi (\vec{e}_z \partial_z \varphi)] \\ &= \partial_{rr} \varphi + \frac{1}{r} \partial_r \varphi + \frac{1}{r^2} \partial_{\phi\phi} \varphi + \partial_{zz} \varphi \\ &= \frac{1}{r} \partial_r (r \partial_r \varphi) + \frac{1}{r^2} \partial_{\phi\phi} \varphi + \partial_{zz} \varphi \end{aligned}$$

7. 球坐标下的导数运算.

$$d\vec{r} = dr\vec{e}_r + r d\vec{e}_\theta + r \sin\theta d\vec{e}_\phi$$

$$dV = r^2 \sin\theta dr d\theta d\phi$$

$$\begin{aligned}\vec{v} &= \vec{e}_r \frac{dr}{dt} + \vec{e}_\theta r \frac{d\theta}{dt} + \vec{e}_\phi r \sin\theta \frac{d\phi}{dt} \\ &= \vec{e}_r \dot{r} + \vec{e}_\theta \frac{1}{r} \dot{\theta} + \vec{e}_\phi \frac{1}{r \sin\theta} \dot{\phi}\end{aligned}$$

$$\vec{v} \cdot \vec{v} = \dot{r}^2 + \dot{\theta}^2 + \dot{\phi}^2$$

$$\frac{\partial \vec{e}_r}{\partial r} = 0, \quad \frac{\partial \vec{e}_\theta}{\partial r} = 0, \quad \frac{\partial \vec{e}_\phi}{\partial r} = 0$$

$$\frac{\partial \vec{e}_r}{\partial \theta} = \vec{e}_\theta, \quad \frac{\partial \vec{e}_\theta}{\partial \theta} = -\vec{e}_r, \quad \frac{\partial \vec{e}_\phi}{\partial \theta} = 0$$

$$\frac{\partial \vec{e}_r}{\partial \phi} = \sin\theta \vec{e}_\phi, \quad \frac{\partial \vec{e}_\theta}{\partial \phi} = \cos\theta \vec{e}_\phi, \quad \frac{\partial \vec{e}_\phi}{\partial \phi} = -\sin\theta \vec{e}_r - \cos\theta \vec{e}_\theta$$

$$\begin{aligned}\vec{v} \cdot \vec{A} &= (\vec{e}_r \dot{r} + \vec{e}_\theta \frac{1}{r} \dot{\theta} + \vec{e}_\phi \frac{1}{r \sin\theta} \dot{\phi}) \cdot (A_r \vec{e}_r + A_\theta \vec{e}_\theta + A_\phi \vec{e}_\phi) \\ &= \vec{e}_r \cdot [\dot{r}(A_r \vec{e}_r) + \dot{\theta}(A_\theta \vec{e}_\theta) + \dot{\phi}(A_\phi \vec{e}_\phi)] + \vec{e}_\theta \cdot \frac{1}{r} [\dot{\theta}(A_r \vec{e}_r) + \dot{\theta}(A_\theta \vec{e}_\theta) + \dot{\phi}(A_\phi \vec{e}_\phi)] \\ &\quad + \vec{e}_\phi \cdot \frac{1}{r \sin\theta} [\dot{\phi}(A_r \vec{e}_r) + \dot{\phi}(A_\theta \vec{e}_\theta) + \dot{\phi}(A_\phi \vec{e}_\phi)] \\ &= \dot{r} A_r + \frac{1}{r} \dot{\theta} A_\theta + \frac{1}{r} \dot{\phi} A_\phi + \frac{1}{r} \cos\theta \dot{\phi} A_\theta + \frac{1}{r \sin\theta} \dot{\phi} A_\phi \\ &= \left(\frac{1}{r^2} + \dot{r} \right) A_r + \left(\frac{1}{r} \dot{\theta} + \frac{1}{r} \cos\theta \dot{\phi} \right) A_\theta + \frac{1}{r \sin\theta} \dot{\phi} A_\phi \\ &= \frac{1}{r^2} \dot{r} (r^2 A_r) + \frac{1}{r \sin\theta} \dot{\theta} (\sin\theta A_\theta) + \frac{1}{r \sin\theta} \dot{\phi} A_\phi\end{aligned}$$

$$\vec{v} \cdot \vec{e}_r = \frac{1}{r^2} \dot{r} (r^2) = \frac{\dot{r}}{r}$$

$$\begin{aligned}\vec{v} \times \vec{A} &= (\vec{e}_r \dot{r} + \vec{e}_\theta \frac{1}{r} \dot{\theta} + \vec{e}_\phi \frac{1}{r \sin\theta} \dot{\phi}) \times (A_r \vec{e}_r + A_\theta \vec{e}_\theta + A_\phi \vec{e}_\phi) \\ &= \vec{e}_r \times [\dot{r}(A_r \vec{e}_r) + \dot{\theta}(A_\theta \vec{e}_\theta) + \dot{\phi}(A_\phi \vec{e}_\phi)] + \vec{e}_\theta \times \frac{1}{r} [\dot{\theta}(A_r \vec{e}_r) + \dot{\theta}(A_\theta \vec{e}_\theta) + \dot{\phi}(A_\phi \vec{e}_\phi)] \\ &\quad + \vec{e}_\phi \times \frac{1}{r \sin\theta} [\dot{\phi}(A_r \vec{e}_r) + \dot{\phi}(A_\theta \vec{e}_\theta) + \dot{\phi}(A_\phi \vec{e}_\phi)] \\ &= \dot{r} A_\theta \vec{e}_\phi - \dot{r} A_\phi \vec{e}_\theta + \frac{1}{r} \dot{\theta} A_r (-\vec{e}_\phi) + \frac{1}{r} A_\theta (\vec{e}_\phi) + \frac{1}{r} \dot{\theta} A_\phi \vec{e}_r \\ &\quad + \frac{1}{r \sin\theta} \dot{\phi} A_r \vec{e}_\theta + \frac{1}{r \sin\theta} \dot{\phi} A_\theta (-\vec{e}_r) + \frac{A_\phi}{r \sin\theta} (-\sin\theta \vec{e}_\theta + \cos\theta \vec{e}_r) \\ &= \left(\frac{1}{r} \dot{\theta} A_\phi - \frac{1}{r \sin\theta} \dot{\phi} A_\theta + \frac{1}{r \sin\theta} A_\phi \cos\theta \right) \vec{e}_r + \left(-\dot{r} A_\phi + \frac{1}{r \sin\theta} \dot{\phi} A_r - \frac{1}{r} A_\phi \right) \vec{e}_\theta \\ &\quad + (\dot{r} A_\theta - \frac{1}{r} \dot{\theta} A_r + \frac{1}{r} A_\theta) \vec{e}_\phi \\ &= \frac{1}{r \sin\theta} [\dot{\theta} (\sin\theta A_\phi) - \dot{\phi} A_\theta] \vec{e}_r + \frac{1}{r} [\dot{\phi} A_r - \dot{r} (r A_\phi)] \vec{e}_\theta \\ &\quad + (\dot{r} A_\theta - \frac{1}{r} \dot{\theta} A_r + \frac{1}{r} A_\theta) \vec{e}_\phi\end{aligned}$$

$$+ \frac{1}{r} [\partial_r (r \partial_\theta) - \partial_\theta \partial_r]$$

$$\nabla^2 \varphi = \nabla \cdot \nabla \varphi$$

$$\begin{aligned} &= (\vec{e}_r \partial_r + \vec{e}_\theta \frac{1}{r} \partial_\theta + \vec{e}_\phi \frac{1}{r \sin \theta} \partial_\phi) \cdot (\vec{e}_r \partial_r \varphi + \vec{e}_\theta \frac{1}{r} \partial_\theta \varphi + \vec{e}_\phi \frac{1}{r \sin \theta} \partial_\phi \varphi) \\ &= \vec{e}_r \cdot \partial_r (\vec{e}_r \partial_r \varphi + \vec{e}_\theta \frac{1}{r} \partial_\theta \varphi + \vec{e}_\phi \frac{1}{r \sin \theta} \partial_\phi \varphi) + \\ &\quad \vec{e}_\theta \cdot \frac{1}{r} \partial_\theta (\vec{e}_r \partial_r \varphi + \vec{e}_\theta \frac{1}{r} \partial_\theta \varphi + \vec{e}_\phi \frac{1}{r \sin \theta} \partial_\phi \varphi) + \\ &\quad \vec{e}_\phi \cdot \frac{1}{r \sin \theta} \partial_\phi (\vec{e}_r \partial_r \varphi + \vec{e}_\theta \frac{1}{r} \partial_\theta \varphi + \vec{e}_\phi \frac{1}{r \sin \theta} \partial_\phi \varphi) \\ &= \partial_r r \varphi + \frac{1}{r} \partial_r \varphi + \frac{1}{r^2} \partial_{\theta\theta} \varphi + \frac{1}{r} \partial_r \varphi + \frac{\cos \theta}{r^2 \sin \theta} \partial_\theta \varphi + \frac{1}{r^2 \sin^2 \theta} \partial_{\phi\phi} \varphi \\ &= \frac{1}{r^2} \partial_r (r^2 \partial_r \varphi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta \varphi) + \frac{1}{r^2 \sin^2 \theta} \partial_{\phi\phi} \varphi \end{aligned}$$

* 一般曲线正交坐标系: (u_1, u_2, u_3)

$$d\vec{r} = du_1 \vec{e}_1 + du_2 \vec{e}_2 + du_3 \vec{e}_3$$

$$= h_1 du_1 \vec{e}_1 + h_2 du_2 \vec{e}_2 + h_3 du_3 \vec{e}_3$$

h_1, h_2, h_3 为标量函数

$$\nabla \varphi = \vec{e}_1 \frac{1}{h_1} \frac{\partial \varphi}{\partial u_1} + \vec{e}_2 \frac{1}{h_2} \frac{\partial \varphi}{\partial u_2} + \vec{e}_3 \frac{1}{h_3} \frac{\partial \varphi}{\partial u_3}$$

$$\nabla \cdot \vec{r} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 \partial_{u_1}) + \frac{\partial}{\partial u_2} (h_1 h_3 \partial_{u_2}) + \frac{\partial}{\partial u_3} (h_1 h_2 \partial_{u_3}) \right]$$

$$\nabla \times \vec{r} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \vec{e}_1 h_1 & \vec{e}_2 h_2 & \vec{e}_3 h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 \partial_{u_1} & h_2 \partial_{u_2} & h_3 \partial_{u_3} \end{vmatrix}$$

$$\nabla^2 \varphi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial}{\partial u_3} \right) \right] \varphi$$

§ 1.3 矢量的积分运算.

1. 梯度积分定理.

$$\int_{\sigma_1}^{\sigma_2} \nabla \varphi \cdot d\vec{r} = \varphi(\vec{r}_2) - \varphi(\vec{r}_1)$$

2. Gauss 定理

$$\int_V \nabla \cdot \vec{A} = \oint_S d\vec{S} \cdot \vec{A}$$

$$\hookrightarrow \int_V \nabla \varphi = \oint_S d\vec{S} \varphi \quad \vec{A} = \varphi \vec{n}$$

$$\hookrightarrow \int_V \nabla \cdot (\vec{A} \times \vec{B}) = \oint_S d\vec{S} \times \vec{A} \quad \vec{A} = \vec{B} \times \vec{n}$$

$$\begin{aligned} \text{证: } \int_V \nabla \cdot (\vec{A} \times \vec{B}) &= \int_V dV [(\nabla \times \vec{A}) \cdot \vec{B} - (\nabla \times \vec{B}) \cdot \vec{A}] \\ &= \int_V \nabla \cdot (\nabla \times \vec{A}) \cdot \vec{B} = \oint_S d\vec{S} \cdot (\vec{A} \times \vec{B}) = \oint_S (d\vec{S} \times \vec{A}) \cdot \vec{B} \quad \square. \end{aligned}$$

$$\text{例: 证明: } \oint_S d\vec{S} = 0$$

$$\text{证: 取 } \vec{A} = \vec{n}$$

$$\text{证明: } V = \frac{1}{3} \oint_S d\vec{S} \cdot \vec{r}$$

$$\text{证: } V = \frac{1}{3} \oint_V dV (\nabla \cdot \vec{r}) = \oint_V dV = 1$$

3. Stokes 定理.

$$\int_S d\vec{S} \cdot (\nabla \times \vec{A}) = \int_C d\vec{l} \cdot \vec{A}$$

$$\hookrightarrow \int_S d\vec{S} \cdot \nabla \varphi = \int_C d\vec{l} \varphi \quad \vec{A} = \varphi \vec{n}$$

$$\hookrightarrow \int_S d\vec{S} \quad \vec{A} = \vec{r} \times \vec{n}$$

$$\text{例: 证明: } \vec{S} = \frac{1}{2} \oint_C \vec{r} \times d\vec{l}$$

$$-\frac{1}{2} \oint_C d\vec{l} \cdot (\vec{r} \times \vec{n}) = -\frac{1}{2} \oint_C (d\vec{l} \times \vec{r}) \cdot \vec{n} \quad n \text{ 为常矢量.}$$

$$= -\frac{1}{2} \int_S d\vec{S} \cdot [\nabla \times (\vec{r} \times \vec{n})] = -\frac{1}{2} \int_S d\vec{S} \cdot [\nabla_r \times (\vec{r} \times \vec{n}) + \nabla_n \times (\vec{r} \times \vec{n})]$$

$$= -\frac{1}{2} \int_S d\vec{S} \cdot [(\vec{n} \cdot \nabla_r) \vec{r} - (\nabla_r \cdot \vec{r}) \vec{n} + (\nabla_n \cdot \vec{n}) \vec{r} - (\vec{r} \cdot \nabla_n) \vec{n}]$$

$$= -\frac{1}{4} \int_S d\vec{S} \left[(\vec{n} \cdot \nabla r) \vec{r} - (\nabla r \cdot \vec{r}) \vec{n} \right] =$$

4. Dirac 函数

$$\int_V dV \nabla \cdot \frac{\vec{r}}{r^3} = \oint_S d\vec{S} \cdot \frac{\vec{r}}{r^3} = 4\pi$$

$$\delta(\vec{r}) = 0 \quad (\vec{r} \notin U), \quad \int_V dV \delta(\vec{r}) = \begin{cases} 1 & \vec{r} \in U \\ 0 & \vec{r} \notin U \end{cases}$$

$$\int dV \varphi(\vec{r}') \delta(\vec{r} - \vec{r}')$$

$$\Leftrightarrow \delta(\vec{r}) = \frac{1}{4\pi} \nabla \cdot \frac{\vec{r}}{r^3} = -\frac{1}{4\pi} \nabla^2 \frac{1}{r} \quad / \quad \nabla^2 \frac{1}{r} = -4\pi \delta(\vec{r})$$

5. Helmholtz 定理

$$R' \rightarrow \infty \text{ 时: } \oint_{S'} d\vec{S}' \cdot \frac{\vec{r}(\vec{R}')}{|\vec{R} - \vec{R}'|}, \quad \oint_{S'} d\vec{S}' \times \frac{\vec{r}(\vec{R}')}{|\vec{R} - \vec{R}'|} \text{ 收敛, } \rightarrow 0$$

$$\text{则 } \vec{F}(\vec{R}) = \nabla \times \vec{A}(\vec{R}) - \nabla \varphi(\vec{R})$$

$$\text{其中 } \varphi(\vec{R}) = \frac{1}{4\pi} \int_V dV' \frac{\nabla' \cdot \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|}$$

$$\vec{A}(\vec{R}) = \frac{1}{4\pi} \int_V dV' \frac{\nabla' \times \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|}$$

$$\text{证: } \vec{F}(\vec{R}) = \int_V dV' \vec{F}(\vec{R}') \delta(\vec{R} - \vec{R}')$$

$$= -\frac{1}{4\pi} \int_V dV' \vec{F}(\vec{R}') \nabla^2 \frac{1}{|\vec{R} - \vec{R}'|} = -\frac{1}{4\pi} \int_V dV' \nabla^2 \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} \quad \nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$$

$$= \frac{1}{4\pi} \int_V dV' \left[\nabla \times (\nabla \times \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|}) - \nabla(\nabla \cdot \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|}) \right]$$

$$\nabla \times \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} = \frac{\nabla \times \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} + \vec{F}(\vec{R}') \nabla \times \frac{1}{|\vec{R} - \vec{R}'|} = \frac{\nabla \times \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|}$$

$$\nabla' \times \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} = \frac{\nabla' \times \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} + \vec{F}(\vec{R}') \nabla' \times \frac{1}{|\vec{R} - \vec{R}'|} = \frac{\nabla' \times \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} - \vec{F}(\vec{R}') \nabla \times \frac{1}{|\vec{R} - \vec{R}'|}$$

$$\nabla \cdot \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} = \frac{\nabla \cdot \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} + \vec{F}(\vec{R}') \nabla \cdot \frac{1}{|\vec{R} - \vec{R}'|} = \frac{\nabla \cdot \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|}$$

$$\nabla' \cdot \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} = \frac{\nabla' \cdot \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} + \vec{F}(\vec{R}') \nabla' \cdot \frac{1}{|\vec{R} - \vec{R}'|} = \frac{\nabla' \cdot \vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|} - \vec{F}(\vec{R}') \nabla \cdot \frac{1}{|\vec{R} - \vec{R}'|}$$

$$\left[\frac{\nabla \times (\nabla \times \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|})}{|\vec{R} - \vec{R}'|} - \frac{\nabla(\nabla \cdot \frac{\vec{F}(\vec{R}')}{|\vec{R} - \vec{R}'|})}{|\vec{R} - \vec{R}'|} \right] = \frac{\nabla \times (\nabla \times \vec{F}(\vec{R}'))}{|\vec{R} - \vec{R}'|} - \frac{\nabla(\nabla \cdot \vec{F}(\vec{R}'))}{|\vec{R} - \vec{R}'|}$$

$$= \frac{1}{4\pi} \int_V dV \left[\nabla \times \left(\frac{\nabla \times \vec{F}(\vec{r}')}{|\vec{r} - \vec{r}'|} \right) - \nabla \left(\frac{\nabla \cdot \vec{F}(\vec{r}')}{|\vec{r} - \vec{r}'|} \right) \right] - \frac{1}{4\pi} \oint_S d\vec{S} \times \left(\nabla \times \frac{\vec{F}(\vec{r}')}{|\vec{r} - \vec{r}'|} \right) - d\vec{S} \cdot \nabla \cdot \frac{\vec{F}(\vec{r}')}{|\vec{r} - \vec{r}'|}$$

$$= -\nabla \varphi(\vec{r}) + \nabla \times \vec{A}(\vec{r})$$

§ 1.4 正交变换

1. 二维转动变换

$$\begin{cases} x' = x \cos \theta + y \sin \theta \\ y' = -x \sin \theta + y \cos \theta \end{cases}$$

2. 三维正交变换

线性: $x'_j = a_{ij} x_i$

正交: $x_i x'_i = x_i x_i$

3. 空间转动

eg. 绕 z 轴 $\vec{a} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

镜面反射 $\vec{a} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (x \rightarrow -x)$

空间反演 $\vec{a} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

4.

可根据物理量在空间变换下的性质分类

标量: $\varphi' = \varphi$

矢量: $A_i' = a_{ij} A_j \quad A_i \vec{e}_i' = A_j \vec{e}_j \quad (\text{转动下, 模长不变})$

5.

1. 真矢量: $\vec{r} \rightarrow -\vec{r}, \vec{p} \rightarrow -\vec{p}$

2. 赝矢量: $\vec{r} \rightarrow -\vec{r}, \vec{p} \rightarrow \vec{p}$

真 $\times$ 真 = 赝

§ 1.5 张量

1. m 维空间中 n 阶张量有 m^n 个分量

二阶张量 $\vec{T}$: $T_{ij} = a_{im} a_{jn} T_{mn}$

2. 并矢, 外积: $\vec{A} \otimes \vec{B} = A_i B_j \vec{e}_i \vec{e}_j \quad * \vec{A} \otimes \vec{B} \neq \vec{B} \otimes \vec{A}$

3. 对称张量: $\vec{T} = T_{ij} \vec{e}_i \vec{e}_j, \quad T_{ij} = T_{ji}$

反对称张量: $T_{ij} = -T_{ji}$

例: $\vec{T}$ 为对称张量, 证明 $\nabla \cdot (\vec{T} \times \vec{r}) = -\vec{r} \times (\nabla \cdot \vec{T})$

证: $\nabla \cdot (T_{ij} \vec{e}_i \vec{e}_j \times r_k \vec{e}_k) = \nabla \cdot (T_{ij} e_l \epsilon_{ijk} e_l r_k)$

又乘元和 $= \epsilon_{lmn} \cdot (e_l e_l T_{ij} \epsilon_{ijk} r_k) = \partial_l (T_{ij} \epsilon_{ijk} r_k) e_l$

前面作用 $= \epsilon_{ijk} [\partial_l T_{ij} r_k + T_{ij} \partial_l r_k] e_l \quad \text{或} \quad \partial_l r_k = \delta_{lk}$

$= \epsilon_{ijk} (\partial_l T_{ij}) r_k e_l + \epsilon_{ijk} T_{ij} e_l = -\epsilon_{ijk} T_{ji} e_l \rightarrow 0$

$= \epsilon_{ijk} (\partial_l T_{ij}) r_k e_l = -\epsilon_{ikj} e_l r_k (\partial_l T_{ij}) = -\vec{r} \times (\nabla \cdot \vec{T})$

单位张量, $\vec{I} = \delta_{ij} \vec{e}_i \vec{e}_j$

例: 证明 $\vec{I} \times \vec{A} = \vec{A} \times \vec{I}, \quad (\vec{A} \times \vec{B}) \times \vec{I} = \vec{B} \otimes \vec{A} - \vec{A} \otimes \vec{B}, \quad \vec{A} \cdot (\vec{I} \times \vec{B}) = \vec{A} \times \vec{B}$

证:

1) $\vec{I} \times \vec{A} = e_i e_l \times A_j e_j = A_j e_l (\epsilon_{kij} e_k)$

$= A_j (\epsilon_{ijk} e_i) e_k = A_j (e_j \times e_k) e_k = \vec{A} \times \vec{I}$

2) $(\vec{A} \times \vec{B}) \times \vec{I} = (\epsilon_{ijk} e_i A_j B_k) \times e_l e_l = \epsilon_{ijk} \epsilon_{lmn} e_m A_j B_k e_l$

$= \epsilon_{ijk} \epsilon_{lilm} e_m A_j B_k e_l = (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) e_m A_j B_k e_l$

$= (B_m e_m) (A_l e_l) - (A_m e_m) (B_l e_l) = \vec{B} \otimes \vec{A} - \vec{A} \otimes \vec{B}$

3) $\vec{A} \cdot (\vec{I} \times \vec{B}) = A_i e_l \cdot (e_j e_j \times B_k e_k)$

$$= A_j e_j \times B_k e_k = \vec{A} \times \vec{B}$$

4. 代数运算

$$(1) (\vec{A} \cdot \vec{B}) \vec{C} = \vec{A} (\vec{B} \cdot \vec{C})$$

$$\vec{C} \cdot (\vec{A} \vec{B}) = (\vec{C} \cdot \vec{A}) \vec{B}$$

$$(\vec{A} \vec{B}) \times \vec{C} = \vec{A} (\vec{B} \times \vec{C})$$

$$\vec{C} \times (\vec{A} \vec{B}) = (\vec{C} \times \vec{A}) \vec{B}$$

$$(2) \vec{r} \cdot \vec{r} = r_j \vec{e}_i \vec{e}_j \cdot r_k \vec{e}_k = r_j \vec{e}_i r_k \delta_{jk} = r_j r_j \vec{e}_i \vec{e}_i$$

$$(3) \text{ 单点积: } (\vec{A} \vec{B}) \cdot (\vec{C} \vec{D}) = \vec{A} (\vec{B} \cdot \vec{C}) \vec{D}$$

$$\text{双点积: } (\vec{A} \vec{B}) \cdot (\vec{C} \vec{D}) = (\vec{B} \cdot \vec{C}) (\vec{A} \cdot \vec{D})$$

$$\text{叉积: } (\vec{A} \vec{B}) \times (\vec{C} \vec{D}) = \vec{A} (\vec{B} \times \vec{C}) \vec{D}$$

5. 导数规则

$$\begin{aligned} \nabla (\varphi \vec{A}) &= \vec{e}_i \partial_i (\varphi A_j \vec{e}_j) = \vec{e}_i \vec{e}_j \left[(\partial_i \varphi) A_j + \varphi (\partial_i A_j) \right] \\ &= (\nabla \varphi) \vec{A} + \varphi (\nabla \vec{A}) \quad \text{并矢} \end{aligned}$$

$$\begin{aligned} \nabla (\vec{A} \cdot \vec{B}) &= \vec{e}_i \partial_i (A_j B_k \delta_{jk}) = \vec{e}_i \left[(\partial_i A_j) B_j + A_j (\partial_i B_j) \right] \\ &= (\nabla \vec{A}) \cdot \vec{B} + (\nabla \vec{B}) \cdot \vec{A} \end{aligned}$$

$$\begin{aligned} \nabla \times (\vec{A} \vec{B}) &= \vec{e}_i \partial_i \times (A_j \vec{e}_j B_k \vec{e}_k) \\ &= \vec{e}_i \times (\partial_i A_j) \vec{e}_j B_k \vec{e}_k + \vec{e}_i \times A_j \vec{e}_j (\partial_i B_k) \vec{e}_k \\ &= \epsilon_{ijk} \vec{e}_i (\partial_i A_j) B_k \vec{e}_k + \epsilon_{ijk} \vec{e}_i A_j (\partial_i B_k) \vec{e}_k \\ &= \epsilon_{ijk} \vec{e}_i (\partial_i A_j) B_k \vec{e}_k + \epsilon_{jik} \vec{e}_i A_j (\partial_i B_k) \vec{e}_k \\ &= (\nabla \times \vec{A}) \vec{B} - (\vec{A} \times \nabla) \vec{B} \end{aligned}$$

$$\begin{aligned} \nabla (\vec{A} \times \vec{B}) &= \vec{e}_i \partial_i \epsilon_{jkl} \vec{e}_j A_k B_l \\ &= \vec{e}_i \epsilon_{jkl} \vec{e}_j \left[(\partial_i A_k) B_l + A_k (\partial_i B_l) \right] \end{aligned}$$

$$\begin{aligned}
&= \epsilon_{jkl} (\partial_j A_k) B_l + \epsilon_{jkl} A_k (\partial_j B_l) \\
&= \epsilon_{jkl} (\partial_j A_k) B_l - \epsilon_{jlk} (\partial_j B_k) A_l \\
&= (\nabla \vec{A}) \times \vec{B} - (\nabla \vec{B}) \times \vec{A}
\end{aligned}$$

$$\begin{aligned}
\nabla \cdot (\vec{A} \vec{B}) &= e_i \partial_i (A_j B_k e_k) \\
&= (\partial_j A_k) B_k e_k + A_k (\partial_j B_k) e_k \\
&= (\nabla \cdot \vec{A}) \vec{B} + \vec{A} \cdot (\nabla \vec{B})
\end{aligned}$$

$$\begin{aligned}
\nabla \cdot (\vec{A} \vec{B} \vec{C}) &= (\nabla A + \nabla B + \nabla C) \cdot \vec{A} \vec{B} \vec{C} \\
&= (\nabla \cdot \vec{A}) \vec{B} \vec{C} + (\nabla B \cdot \vec{A}) \vec{B} \vec{C} + (\nabla C \cdot \vec{A}) \vec{B} \vec{C} \\
&= (\nabla \cdot \vec{A}) \vec{B} \vec{C} + (\vec{A} \cdot \nabla) \vec{B} \vec{C} + \vec{B} (\vec{A} \cdot \nabla) \vec{C}
\end{aligned}$$

$$\nabla \vec{r} = e_i \partial_i r_j e_j = \delta_{ij} e_i e_j = \vec{I}$$

$$\nabla \frac{\vec{r}}{r} = \frac{\nabla \vec{r}}{r} - \frac{\nabla r}{r^2} \vec{r} = \frac{1}{r} \vec{I} - \frac{\vec{r} \vec{r}}{r^3}$$

$$\nabla r = e_i \partial_i r_j = \frac{\vec{r}}{r} \quad ?$$

$$\nabla \nabla \frac{1}{r} = -\nabla \left(\frac{\nabla r}{r^2} \right) = -\nabla \frac{\vec{r}}{r^3} = \frac{3 \nabla r}{r^4} \vec{r} - \frac{\nabla \vec{r}}{r^3} = \frac{3 \vec{r} \vec{r}}{r^5} - \frac{\vec{I}}{r^3}$$

$$\begin{aligned}
\vec{I} : \nabla \nabla &= \delta_{ij} e_i e_j : e_k \partial_k e_l \partial_l \\
&= \delta_{ij} (e_j \cdot e_k \partial_k) (e_l \cdot e_l \partial_l) \\
&= \delta_{ij} \partial_j \partial_i = \nabla \cdot \nabla
\end{aligned}$$

6. Gauss 定理和 Stokes 定理,

$$\int_V dV \nabla \cdot \vec{T} = \oint_S d\vec{S} \cdot \vec{T}$$

$$\int_S d\vec{S} \nabla \times \vec{T} = \oint_L d\vec{r} \cdot \vec{T}$$

1) 矢量场 $\vec{T}$ 在 V 的边界面上满足 $\vec{n} \cdot \vec{T} = 0$,

$$2) \int_V \vec{T} dV = - \int_V dV (\nabla \cdot \vec{T}) \vec{r}$$

$$\int_V dV \vec{T} \vec{r} + \int_V dV \vec{r} \vec{T} = - \int_V dV (\nabla \cdot \vec{T}) \vec{r} \vec{r}$$

3) ,

$$A : (\vec{T} \cdot \nabla) \vec{r} = (T_1 \frac{\partial}{\partial x_1} + T_2 \frac{\partial}{\partial x_2} + T_3 \frac{\partial}{\partial x_3}) \cdot (x_1, x_2, x_3)$$

$$1) \quad \nabla \cdot (\vec{J} \vec{r}) = (\nabla \cdot \vec{J}) \vec{r} + \underbrace{(\vec{J} \cdot \nabla) \vec{r}}_{= (\vec{J}_1, \vec{J}_2, \vec{J}_3) =}$$

$$\begin{aligned} \int_V dV \nabla \cdot (\vec{J} \vec{r}) &= \int_V dV (\nabla \cdot \vec{J}) \vec{r} + \int_V dV \vec{J} \\ &= \oint_S d\vec{S} \cdot \vec{J} \vec{r} \\ &= \oint_S d\vec{S} (\vec{n} \cdot \vec{J}) \vec{r} = 0 \end{aligned}$$

$$\begin{aligned} 2) \quad \nabla \cdot (\vec{J} \vec{r} \vec{r}) &= (\nabla \cdot \vec{J}) \vec{r} \vec{r} + (\vec{J} \cdot \nabla) \vec{r} \vec{r} + \vec{r} (\vec{J} \cdot \nabla) \vec{r} \\ &= (\nabla \cdot \vec{J}) \vec{r} \vec{r} + \vec{J} \vec{r} + \vec{r} \vec{J} \end{aligned}$$

$$\begin{aligned} \int_V dV \nabla \cdot (\vec{J} \vec{r} \vec{r}) &= \int_V dV \{ (\nabla \cdot \vec{J}) \vec{r} \vec{r} + \vec{J} \vec{r} + \vec{r} \vec{J} \} \\ &= \oint_S d\vec{S} \cdot \vec{J} \vec{r} \vec{r} = \oint_S d\vec{S} (\vec{n} \cdot \vec{J}) \vec{r} \vec{r} = 0 \end{aligned}$$